

8. (a) Compound **X** is a hydrocarbon that contains 85.6% of carbon by mass. The relative molecular mass of **X** is in the range 50-60.

11.2g of compound **X** decolourised exactly 32.0g of bromine in the absence of light. Further bromine was decolourised when the reaction mixture was placed in direct sunlight.

- (i) Find the empirical formula of **X** and hence its molecular formula. Show clearly how you carried out your calculation. [3]

Molecular formula

- (ii) Explain what can be deduced from the fact that 11.2g of **X** reacts with exactly 32.0g of bromine. [2]

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- (iii) What type of reaction mechanism occurs when **X** reacts with bromine in the absence of light? [1]

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- (iv) What type of reaction mechanism occurs when **X** reacts with more bromine in the presence of sunlight? [1]

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(v) Draw displayed formulae for each of the following.

[3]

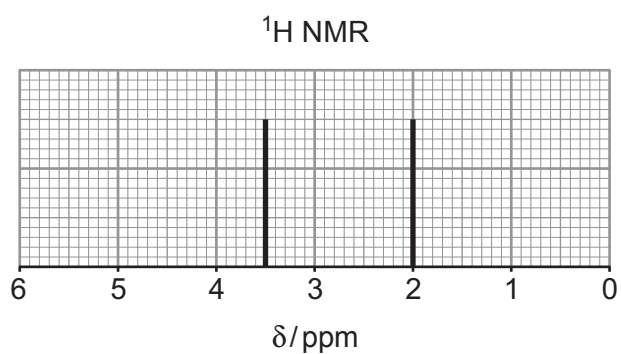
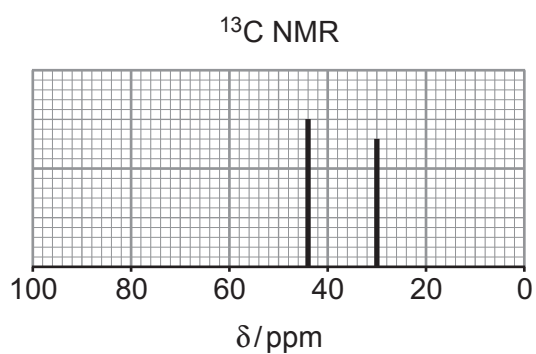
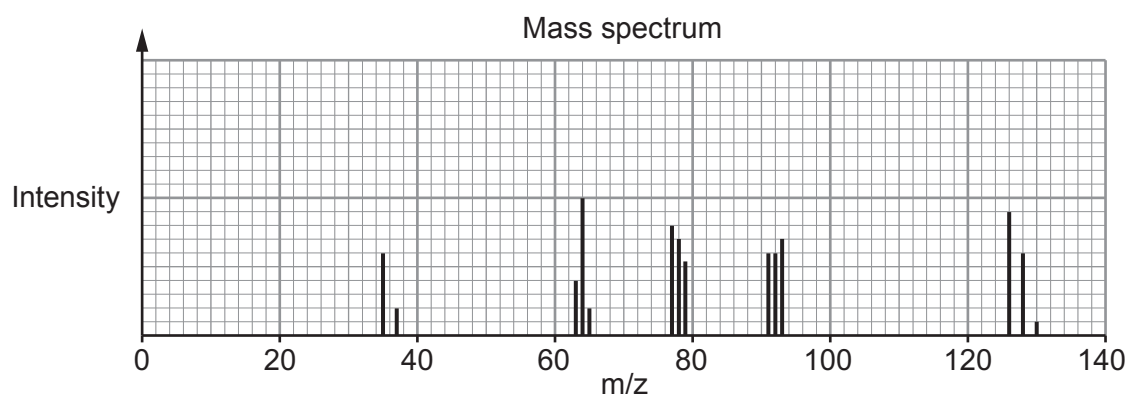
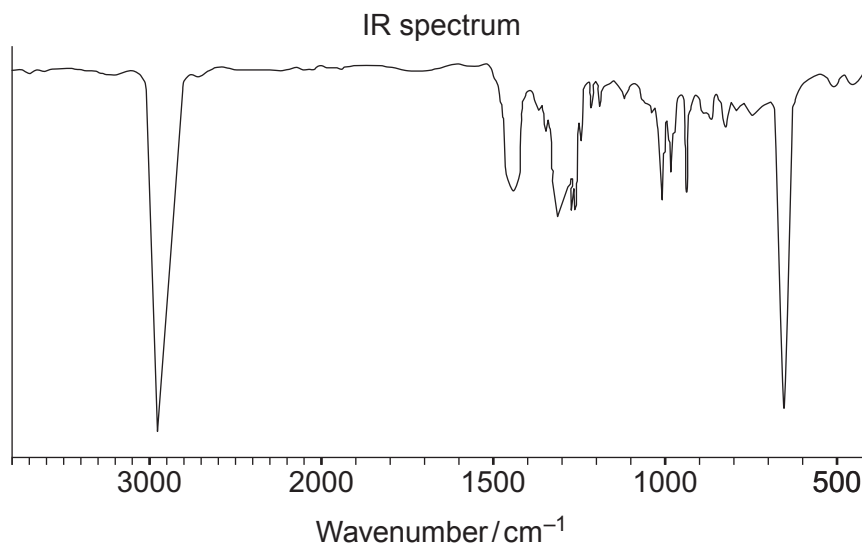
Compound **X**

The product formed when **X** reacts with bromine in the absence of light

One possible product formed when **X** reacts with excess bromine in the presence of sunlight



- (b) Compound **Y** is a halogenoalkane. A simplified form of the IR spectrum, the mass spectrum, the ^{13}C NMR spectrum and the low resolution ^1H NMR spectrum of **Y** are shown below.



Use these spectra to determine the identity of compound **Y**. You must use evidence from each spectrum and explain how you used it. [6 QER]

Examiner
only

